

Literature Benchmark Brief: Dowell_2015_TSG

Title: Very short-term probabilistic wind power forecasts by sparse vector autoregression

Authors: J. Dowell, P. Pinson (2015)

Venue: IEEE Transactions on Smart Grid | **DOI:** 10.1109/TSG.2015.2424078

Cluster: Cluster D: Probabilistic Sizing & Reserve

Core Theoretical Mechanism & Model Formulation

Sparse VAR, generalized logit-normal bounded transformation, pinball loss evaluation

Empirical Scale & Experimental Dataset

22 wind farms in Australia, 5-minute SCADA resolution

Relevance to IEEE TSTE & Sustainable Energy Systems

Demonstrates why symmetric L2 loss is invalid in bounded, asymmetric reserve operations

Identified Gaps & Operational Vulnerabilities

Linear sparse VAR cannot capture non-linear aerodynamic pitch transitions

Role in Manuscript Defense & Reviewer #2 Proof

Theoretical justification for Newsvendor critical fractile $q^* = 1 - 1/\rho$ and Pinball loss